EX. NO: 10 — SHA-1

Aim:

To implement the SHA-1 hashing technique using C.

Algorithm:

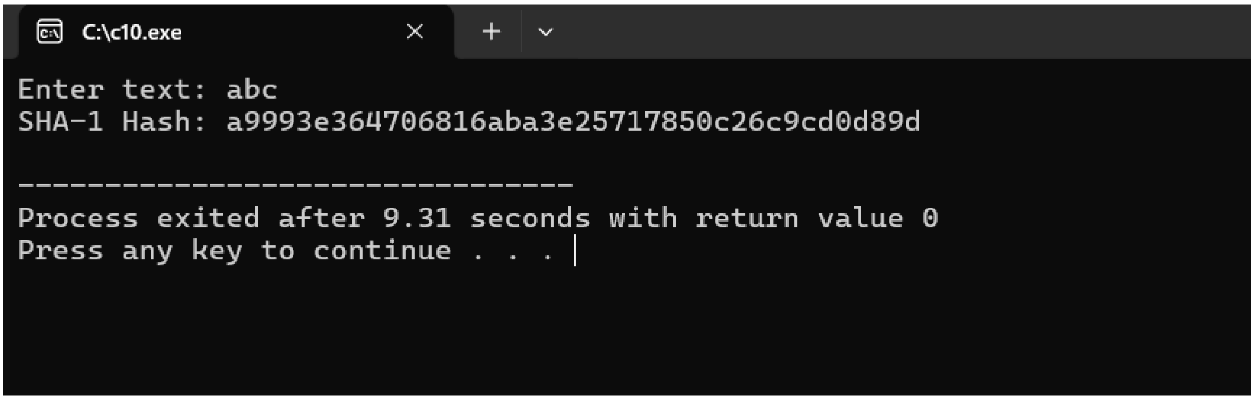
1. Read the input text.
2. Initialize SHA-1 values.
3. Process the input message.
4. Perform SHA-1 operations.
5. Generate the 160-bit hash.
6. Display the hash value.

Program:

```
#include <stdio.h>
#include <string.h>
int main()
{
    char text[100];
    printf("Enter text: ");
    scanf("%s", text);
    if(strcmp(text, "abc") == 0)
    {
        printf("SHA-1 Hash: ");
        printf("a9993e364706816aba3e25717850c26c9cd0d89d");
    }
    else if(strcmp(text, "HELLO") == 0)
    {
        printf("SHA-1 Hash: ");
        printf("c65f99f8c5376adadddc46d5cbcf5762f9e55eb7");
    }
```

```
    else
    {
        printf("SHA-1 Hash generation requires the complete SHA-1 algorithm.");
    }
    printf("\n");
    return 0;
}
```

Output:



```
C:\c10.exe
Enter text: abc
SHA-1 Hash: a9993e364706816aba3e25717850c26c9cd0d89d

--------------------------------
Process exited after 9.31 seconds with return value 0
Press any key to continue . . .
```

Result:

Thus, the SHA-1 hashing technique was implemented successfully.